

# AutoML.

## Часть 2

Юлдуз Фаттахова

Tech Lead, Sber

Skillbox

# Цели модуля

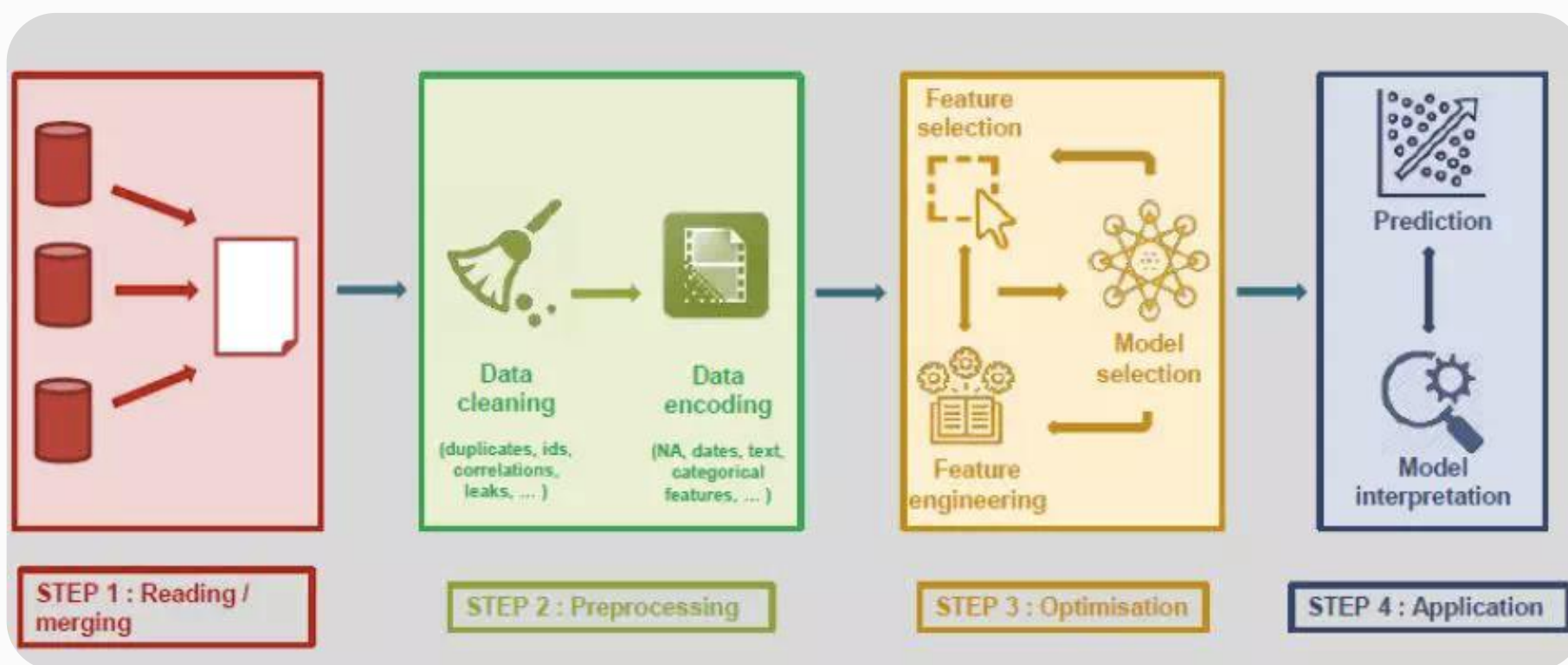
- ✓ Познакомиться с разнообразными библиотеками AutoML для подбора алгоритмов моделирования
- ✓ Разобрать, как устроены AutoML-библиотеки по подбору моделей изнутри
- ✓ Попрактиковаться в применении библиотек автоматизированного машинного обучения

# Библиотеки для работы с AutoML

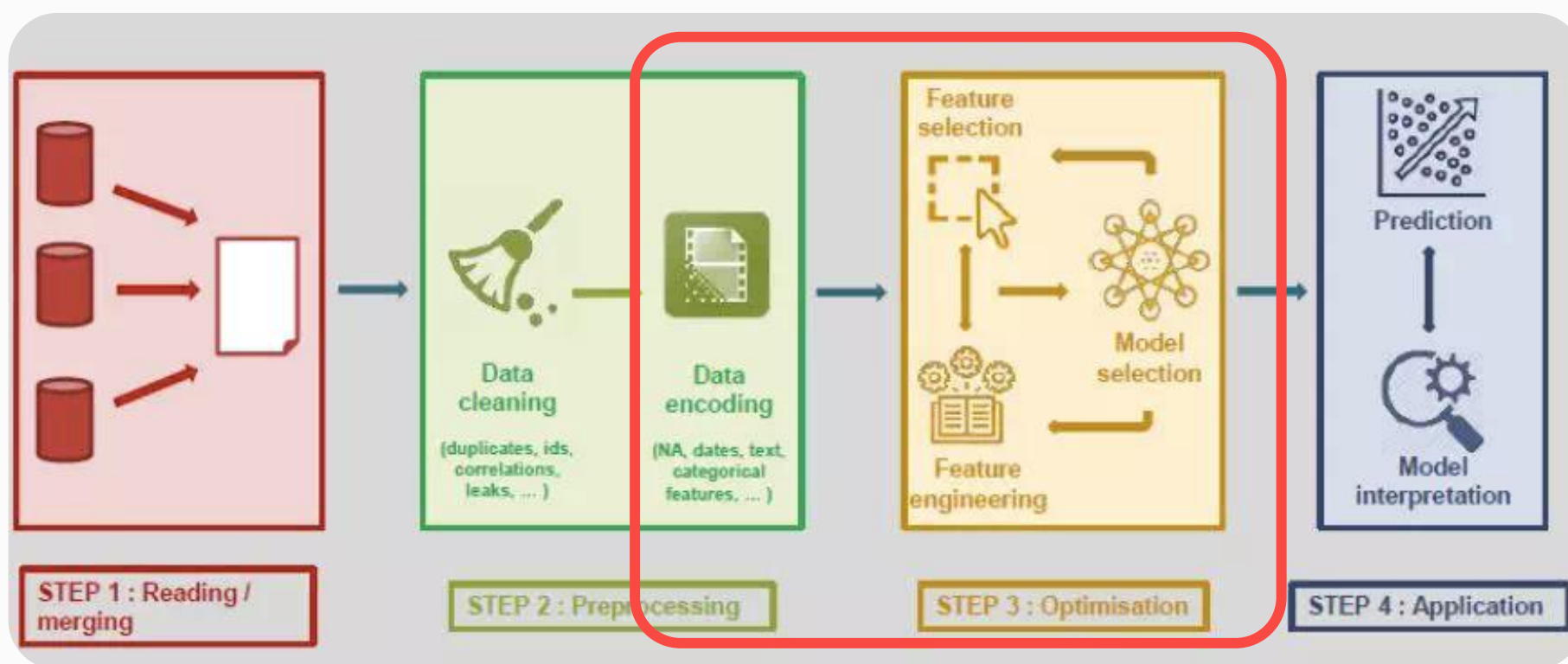
# Цели видео

- ✓ Познакомиться с разнообразными библиотеками AutoML для подбора алгоритмов моделирования
- ✓ Разобраться, по каким критериям выбирать AutoML-инструменты

# Основные этапы решения ML-задачи



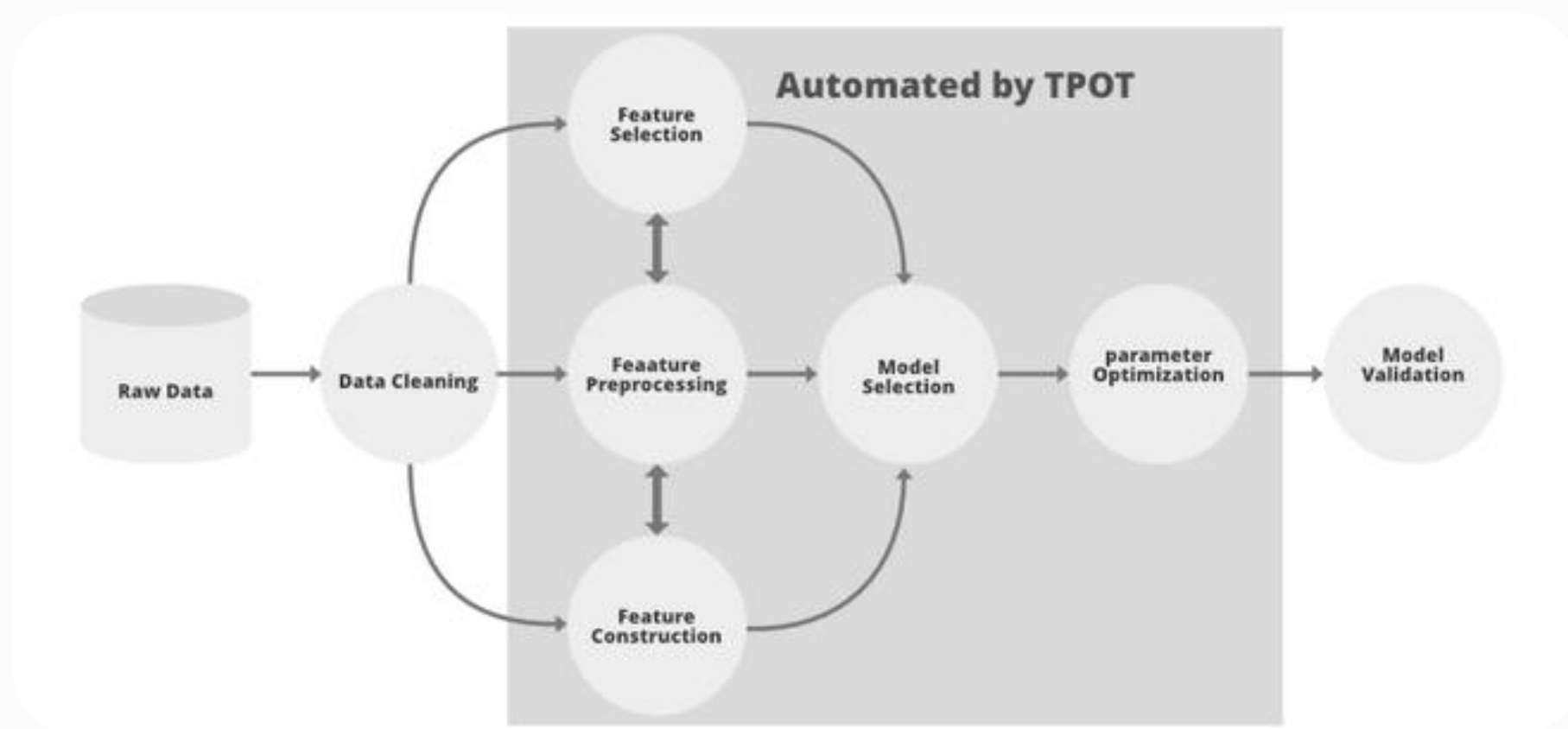
# Основные этапы решения ML-задачи



# Основные задачи инструмента

- ✓ Data Encoding
- ✓ Feature Selection
- ✓ Model Selection
- ✓ Hyperparameter Optimization

# Пример AutoML-инструмента — TPOT





# Основная идея работы с AutoML-инструментом

```
1 from framework import AutoML
2
3 automl = AutoML(time_budget="done by lunch")
4
5 automl.fit(X_train, y_train)
6
7 automl.predict(X_test)
8
9
```

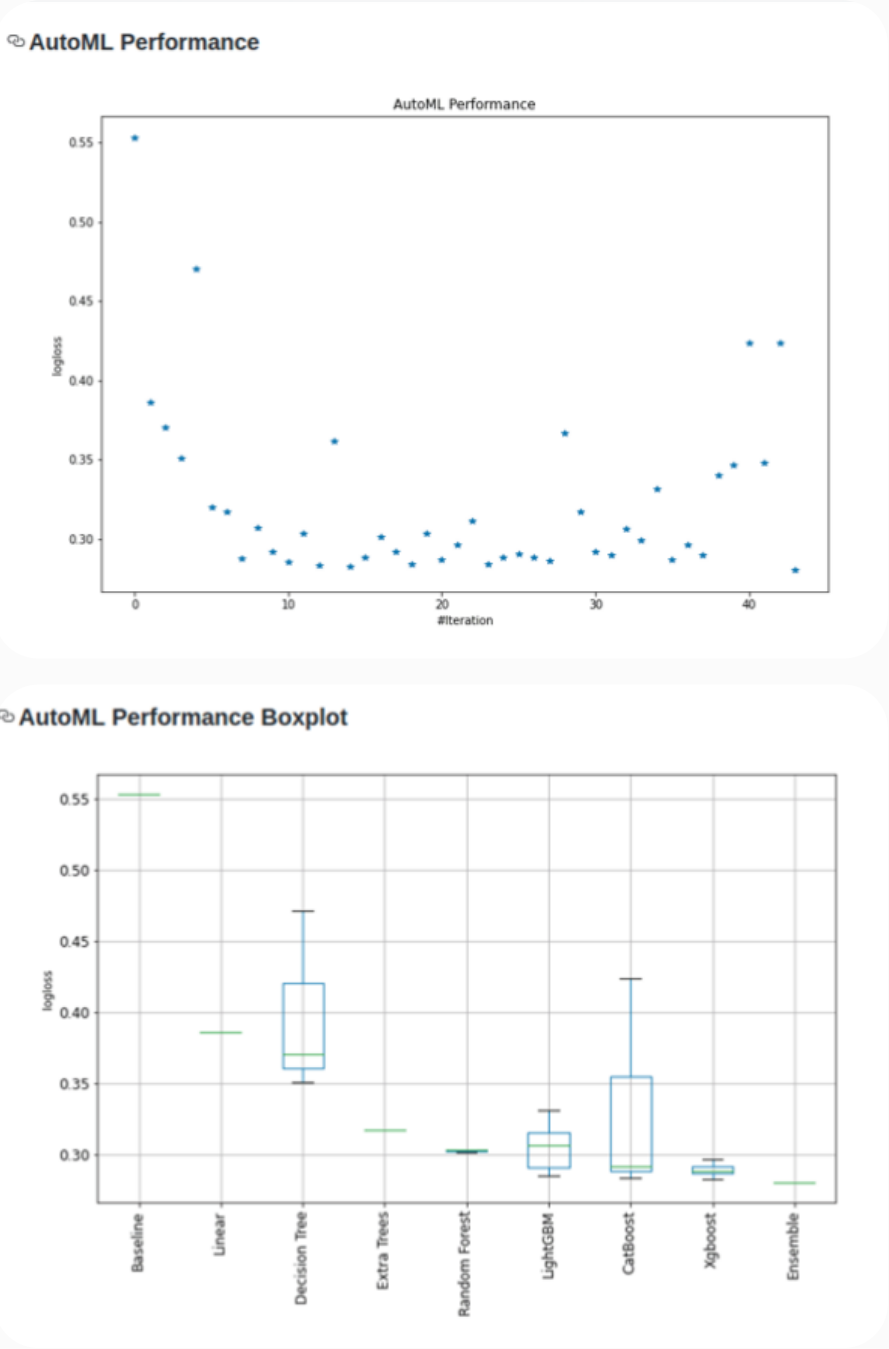
# Зачем использовать AutoML для подбора модели?

**Для упрощения работы ML-инженера и Data Scientist:**

- не нужно писать много однообразного кода
- удобный понятный интерфейс работы с библиотекой
- все метрики качества на разных семействах алгоритмов хранятся в одном месте

# Пример AutoML-отчётов

| AutoML Leaderboard |          |               |             |              |            |                              |
|--------------------|----------|---------------|-------------|--------------|------------|------------------------------|
| Best model         | name     | model_type    | metric_type | metric_value | train_time | Link                         |
|                    | model_1  | Baseline      | logloss     | 0.553216     | 0.57       | <a href="#">Results link</a> |
|                    | model_2  | Linear        | logloss     | 0.386093     | 16.86      | <a href="#">Results link</a> |
|                    | model_3  | Decision Tree | logloss     | 0.37051      | 32.1       | <a href="#">Results link</a> |
|                    | model_4  | Decision Tree | logloss     | 0.35068      | 38.31      | <a href="#">Results link</a> |
|                    | model_5  | Decision Tree | logloss     | 0.470718     | 25.67      | <a href="#">Results link</a> |
|                    | model_35 | LightGBM      | logloss     | 0.319996     | 89.21      | <a href="#">Results link</a> |
|                    | model_24 | Extra Trees   | logloss     | 0.316764     | 758.36     | <a href="#">Results link</a> |
|                    | model_47 | CatBoost      | logloss     | 0.287799     | 43.44      | <a href="#">Results link</a> |
|                    | model_28 | LightGBM      | logloss     | 0.306792     | 95.62      | <a href="#">Results link</a> |
|                    | model_54 | CatBoost      | logloss     | 0.291674     | 43.83      | <a href="#">Results link</a> |
|                    | model_32 | LightGBM      | logloss     | 0.285343     | 37.26      | <a href="#">Results link</a> |
|                    | model_15 | Random Forest | logloss     | 0.303293     | 260.59     | <a href="#">Results link</a> |
|                    | model_44 | Xgboost       | logloss     | 0.282967     | 64.23      | <a href="#">Results link</a> |
|                    | model_51 | CatBoost      | logloss     | 0.361852     | 43.54      | <a href="#">Results link</a> |
|                    | model_36 | Xgboost       | logloss     | 0.282682     | 42.8       | <a href="#">Results link</a> |
|                    | model_42 | Xgboost       | logloss     | 0.288004     | 61.22      | <a href="#">Results link</a> |
|                    | model_8  | Random Forest | logloss     | 0.30142      | 261.95     | <a href="#">Results link</a> |
|                    | model_37 | Xgboost       | logloss     | 0.29182      | 38.75      | <a href="#">Results link</a> |
|                    | model_49 | CatBoost      | logloss     | 0.28389      | 31.79      | <a href="#">Results link</a> |
|                    | model_11 | Random Forest | logloss     | 0.303303     | 200.62     | <a href="#">Results link</a> |
|                    | model_53 | CatBoost      | logloss     | 0.286871     | 43.73      | <a href="#">Results link</a> |
|                    | model_39 | Xgboost       | logloss     | 0.296138     | 45.37      | <a href="#">Results link</a> |
|                    | model_27 | LightGBM      | logloss     | 0.311163     | 68.87      | <a href="#">Results link</a> |
|                    | model_55 | CatBoost      | logloss     | 0.290329     | 43.68      | <a href="#">Results link</a> |
|                    | model_31 | LightGBM      | logloss     | 0.287866     | 50.21      | <a href="#">Results link</a> |
|                    | model_41 | Xgboost       | logloss     | 0.28635      | 44.36      | <a href="#">Results link</a> |
|                    | model_48 | CatBoost      | logloss     | 0.366781     | 43.65      | <a href="#">Results link</a> |
|                    | model_29 | LightGBM      | logloss     | 0.31711      | 76.78      | <a href="#">Results link</a> |
|                    | model_38 | Xgboost       | logloss     | 0.291679     | 41.4       | <a href="#">Results link</a> |
|                    | model_46 | CatBoost      | logloss     | 0.289934     | 43.92      | <a href="#">Results link</a> |
|                    | model_30 | LightGBM      | logloss     | 0.30651      | 152.52     | <a href="#">Results link</a> |
|                    | model_33 | LightGBM      | logloss     | 0.298923     | 90.55      | <a href="#">Results link</a> |
|                    | model_26 | LightGBM      | logloss     | 0.331075     | 88.7       | <a href="#">Results link</a> |
|                    | model_45 | Xgboost       | logloss     | 0.286455     | 91.6       | <a href="#">Results link</a> |
|                    | model_43 | Xgboost       | logloss     | 0.29637      | 82.26      | <a href="#">Results link</a> |
|                    | model_50 | CatBoost      | logloss     | 0.289398     | 43.63      | <a href="#">Results link</a> |
|                    | model_56 | CatBoost      | logloss     | 0.340194     | 27.88      | <a href="#">Results link</a> |
|                    | model_57 | CatBoost      | logloss     | 0.346185     | 10.35      | <a href="#">Results link</a> |
|                    | model_58 | CatBoost      | logloss     | 0.423313     | 10.47      | <a href="#">Results link</a> |
|                    | model_59 | CatBoost      | logloss     | 0.347703     | 3.38       | <a href="#">Results link</a> |
|                    | model_60 | CatBoost      | logloss     | 0.423381     | 3.53       | <a href="#">Results link</a> |
| *** the best ***   | ensemble | Ensemble      | logloss     | 0.280257     | 18.81      | <a href="#">Results link</a> |



# Какие задачи решают?

Чаще всего AutoML-инструменты моделирования решают задачи обучения с учителем.

# Open Source vs Proprietary



# Как выбрать инструмент?

|            |                           | Auto-Keras | H2O-3 AutoML | LAMA | FLAML | PYCARET | Auto-Gluon | MLJAR-supervised | Ludwig | TPOT | Auto-Sklearn | Auto-PyTorch | GAMA | Hyperopt-sklearn | Auto-WEKA | RECIPE | TransmogriAI | MLBox | Auto_ML |
|------------|---------------------------|------------|--------------|------|-------|---------|------------|------------------|--------|------|--------------|--------------|------|------------------|-----------|--------|--------------|-------|---------|
| Data Types | Structured (Tabular)      | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | YES    | YES          | YES   | YES     |
|            | Images                    | YES        | NO           | YES  | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Text                      | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | YES          | NO    | NO      |
|            | Time-Series               | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | YES          | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Modal               | YES        | NO           | NO   | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
| Target     | Binary Classification     | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | YES    | YES          | YES   | YES     |
|            | MultiClass Classification | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       |        | YES          | YES   | YES     |
|            | Regression                | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | NO     | YES          | YES   | YES     |
|            | Multi-Task                | YES        | NO           | NO   | NO    | NO      | NO         | NO               | YES    | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Label               | NO         | NO           | NO   |       | NO      | NO         | NO               | NO     | NO   | YES          | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Clustering                | NO         | NO           | NO   | NO    | YES     | NO         | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |

Изображение: Urbanowicz, R.~J. and 21 colleagues 2023.\ STREAMLINE: An Automated Machine Learning Pipeline for Biomedicine Applied to Examine the Utility of Photography-Based Phenotypes for OSA Prediction Across International Sleep Centers / [arxiv.org/abs](https://arxiv.org/abs/2305.12345)



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|            |                           | Auto-Keras | H2O-3 AutoML | LAMA | FLAML | PYCARET | Auto-Gluon | MLJAR-supervised | Ludwig | TPOT | Auto-Sklearn | Auto-PyTorch | GAMA | Hyperopt-sklearn | Auto-WEKA | RECIPE | TransmogriAI | MLBox | Auto_ML |
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|            | Images                    | YES        | NO           | YES  | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Text                      | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | YES          | NO    | NO      |
|            | Time-Series               | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | YES          | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Modal               | YES        | NO           | NO   | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
| Target     | Binary Classification     | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | YES    | YES          | YES   | YES     |
|            | MultiClass Classification | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       |        | YES          | YES   | YES     |
|            | Regression                | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | NO     | YES          | YES   | YES     |
|            | Multi-Task                | YES        | NO           | NO   | NO    | NO      | NO         | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Label               | NO         | NO           | NO   |       | NO      | NO         | NO               | NO     | NO   | YES          | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Clustering                | NO         | NO           | NO   | NO    | YES     | NO         | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |

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|            | Text                      | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | YES          | NO    | NO      |
|            | Time-Series               | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | YES          | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Modal               | YES        | NO           | NO   | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
| Target     | Binary Classification     | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | YES    | YES          | YES   | YES     |
|            | MultiClass Classification | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       |        | YES          | YES   | YES     |
|            | Regression                | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | NO     | YES          | YES   | YES     |
|            | Multi-Task                | YES        | NO           | NO   | NO    | NO      | NO         | NO               | YES    | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Label               | NO         | NO           | NO   |       | NO      | NO         | NO               | NO     | NO   | YES          | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Clustering                | NO         | NO           | NO   | NO    | YES     | NO         | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |

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|            | Images                    | YES        | NO           | YES  | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Text                      | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | YES          | NO    | NO      |
|            | Time-Series               | YES        | NO           | YES  | YES   | NO      | YES        | YES              | NO     | NO   | NO           | YES          | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Modal               | YES        | NO           | NO   | NO    | NO      | YES        | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
| Target     | Binary Classification     | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | YES    | YES          | YES   | YES     |
|            | MultiClass Classification | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       |        | YES          | YES   | YES     |
|            | Regression                | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | NO     | YES          | YES   | YES     |
|            | Multi-Task                | YES        | NO           | NO   | NO    | NO      | NO         | NO               | YES    | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Multi-Label               | NO         | NO           | NO   |       | NO      | NO         | NO               | NO     | NO   | YES          | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|            | Clustering                | NO         | NO           | NO   | NO    | YES     | NO         | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |

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# PyCaret

| Name      |                                  | Reference                                         |
|-----------|----------------------------------|---------------------------------------------------|
| ID        |                                  |                                                   |
| kmeans    | K-Means Clustering               | sklearn.cluster._kmeans.KMeans                    |
| ap        | Affinity Propagation             | sklearn.cluster._affinity_propagation.Affinity... |
| meanshift | Mean Shift Clustering            | sklearn.cluster._mean_shift.MeanShift             |
| sc        | Spectral Clustering              | sklearn.cluster._spectral.SpectralClustering      |
| hclust    | Agglomerative Clustering         | sklearn.cluster._agglomerative.AgglomerativeCl... |
| dbscan    | Density-Based Spatial Clustering | sklearn.cluster._dbscan.DBSCAN                    |
| optics    | OPTICS Clustering                | sklearn.cluster._optics.OPTICS                    |
| birch     | Birch Clustering                 | sklearn.cluster._birch.Birch                      |
| kmodes    | K-Modes Clustering               | kmodes.kmodes.KModes                              |

# Как выбрать инструмент?

|                             |                            | Auto-Keras | H2O-3 AutoML | LAMA | FLAML | PyCaret | Auto-Gluon | MLJAR-supervised | Ludwig | TPOT | Auto-Sklearn | Auto-PyTorch | GAMA | Hyperopt-sklearn | Auto-WEKA | RECIPE | TransmogriAI | MLBox | Auto_ML |
|-----------------------------|----------------------------|------------|--------------|------|-------|---------|------------|------------------|--------|------|--------------|--------------|------|------------------|-----------|--------|--------------|-------|---------|
| Ease of Use                 | Codeless Implementation    | YES        | YES          | NO   | NO    | NO      | NO         | YES              | NO     | YES  | YES          | NO           | NO   | NO               | YES       | NO     | NO           | NO    | NO      |
|                             | Codeless Running           | NO         | YES          | NO   | NO    | NO      | NO         | YES              | NO     | NO   | NO           | NO           | NO   | NO               | YES       | NO     | NO           | NO    | NO      |
|                             | GUI                        | NO         | YES          | NO   | NO    | NO      | NO         | YES              | NO     | NO   | NO           | NO           | YES  | NO               | YES       | NO     | NO           | NO    | NO      |
|                             | Generates Figures/Viz      | NO         | NO           | YES  | NO    | YES     | NO         | YES              | YES    | NO   | NO           | NO           | YES  | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Outputs/Oranizes Results   | NO         | NO           | NO   | NO    | NO      | NO         | YES              | YES    | YES  | NO           | NO           | YES  | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Summary Report             | NO         | SOME         | YES  | NO    | YES     | NO         | YES              | YES    | NO   | NO           | NO           | NO   | NO               | NO        | NO     | SOME         | NO    | NO      |
|                             | Recommender System         | NO         | NO           | NO   | NO    | NO      | NO         | NO               | NO     | NO   | NO           | YES          | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Parallizable               | YES        | SOME         | NO   | YES   | SOME    | YES        | NO               | YES    | YES  | YES          | YES          | SOME | NO               | NO        | YES    | YES          | NO    | NO      |
| Automated Pipeline Elements | Pipeline Config. Discovery |            | NO           | NO   | NO    | NO      | NO         | NO               | NO     | YES  | YES          |              | YES  | NO               | NO        | YES    | NO           | NO    | NO      |
|                             | Feature Extraction         | N/A        | NO           | YES  |       | NO      |            | YES              | NO     | NO   | NO           | N/A          | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Exploratory Data Analysis  | NO         | NO           | NO   | NO    | YES     | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Categorical Encoding       | NO         | NO           | YES  | NO    | YES     | YES        | YES              | YES    | NO   | YES          | YES          | NO   | NO               | YES       | NO     | YES          | NO    |         |
|                             | Numerical Encoding         | NO         | NO           | YES  | NO    | YES     | YES        | YES              | YES    | NO   | NO           | NO           | NO   | NO               |           | NO     | NO           |       | YES     |
|                             | Basic Data Cleaning        | NO         | NO           | NO   | NO    | YES     | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | YES   | YES     |
|                             | Missing Val. Imputation    | NO         | NO           | NO   | NO    | YES     | YES        | YES              | YES    | YES  | YES          | YES          | NO   | NO               | NO        | YES    | YES          | NO    |         |
|                             | Feature Engineering        | NO         | NO           | YES  | NO    | YES     | YES        | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | YES    | YES          | NO    | YES     |
|                             | Feature Transformation     | NO         | NO           | NO   | NO    | YES     | NO         | YES              | YES    | YES  | YES          | YES          | YES  | NO               | NO        | NO     | YES          | NO    | YES     |
|                             | Feature Learning           | NO         | NO           | NO   | NO    | NO      | NO         | YES              | NO     | YES  | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | YES     |
|                             | Feature Selection          | NO         | NO           | YES  | NO    | YES     | NO         | YES              | NO     | YES  | YES          | NO           | NO   | NO               | NO        | YES    | YES          | YES   |         |
|                             | Algorithm Selection        | NO         | YES          | SOME | YES   | YES     | YES        | YES              | NO     | YES  | YES          | NO           | YES  | NO               | YES       | YES    | YES          | YES   | YES     |
|                             | Hyperparam. Optimization   | YES        | YES          | YES  | YES   | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | YES              | YES       | YES    | YES          |       | NO      |
|                             | Deep Learning Included     | ONLY       | YES          | NO   |       | NO      | NO         | NO               | SOME   | SOME | NO           | ONLY         | NO   | NO               | NO        | NO     | NO           | YES   | YES     |
|                             | Models for each CV Part.   | NO         | NO           | NO   | NO    | YES     | NO         | YES              | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           |       | NO      |
|                             | Model Evaluation           | NO         | YES          | YES  | NO    | YES     | YES        | YES              | YES    | YES  | YES          | YES          | YES  | NO               | NO        | NO     | YES          | NO    | NO      |
|                             | Ensembling                 | NO         | YES          | YES  | YES   | NO      | YES        | YES              | NO     | NO   | YES          | YES          | YES  | NO               | NO        | NO     | NO           | YES   | NO      |
|                             | Significance Comparisons   | NO         | NO           | NO   | NO    | NO      | NO         | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Directly Compare Datasets  | NO         | NO           | NO   | NO    | NO      | NO         | NO               | NO     | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Replication Analysis       | NO         | YES          | NO   | NO    | NO      |            |                  |        | NO   | NO           |              | NO   | NO               | NO        | NO     | NO           | NO    | NO      |
|                             | Model Deploy (Predictions) | NO         | NO           | NO   | NO    | YES     | NO         | NO               | YES    | NO   | NO           | NO           | NO   | NO               | NO        | NO     | NO           | YES   | NO      |

Изображение: Urbanowicz, R.~J. and 21 colleagues 2023.\ STREAMLINE: An Automated Machine Learning Pipeline for Biomedicine Applied to Examine the Utility of Photography-Based Phenotypes for OSA Prediction Across International Sleep Centers / [arxiv.org/abs](https://arxiv.org/abs/2305.12345)

# Интерпретируемые модели

## **Black Box**

Внутреннюю работу этих моделей сложнее понять: neural networks, gradient boosting models или сложные ансамбли.

## **White Box**

Более простые модели, такие как линейная регрессия и деревья решений, обеспечивают меньшую предсказательную способность и не всегда способны моделировать присущую набору данных сложность (т. е. взаимодействие объектов). Однако, их значительно проще объяснить и интерпретировать.

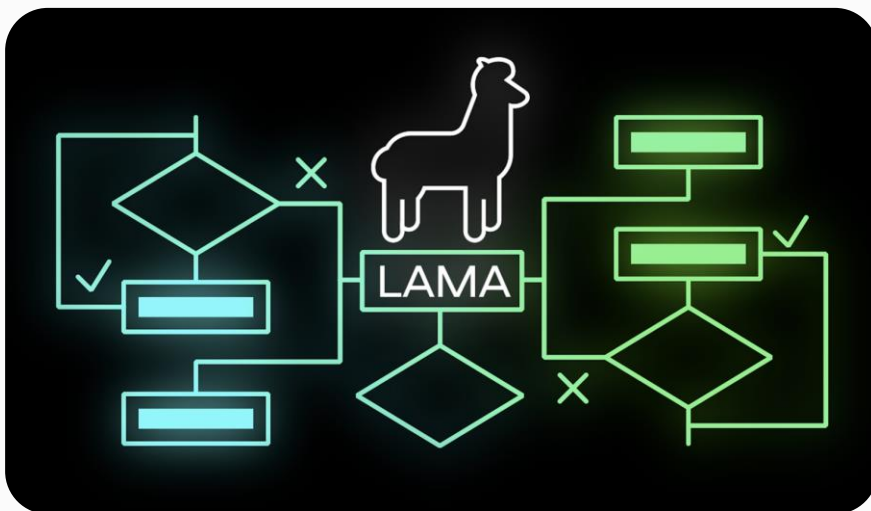
Библиотеки для работы с AutoML

The logo for H2O.ai, featuring the text "H2O.ai" in a bold, black, sans-serif font centered on a solid yellow rounded square background.

**H2O.ai**

The logo for mljar, featuring the word "mljar" in a large, blue, lowercase sans-serif font. Below it, the tagline "Machine Learning for Humans" is written in a smaller, blue, sans-serif font.

**mljar**  
Machine Learning for Humans

The logo for AutoGluon, featuring a blue circular icon with a stylized white 'A' on the left, followed by the text "AutoGluon" in a blue, sans-serif font.

**AutoGluon**



# Библиотеки для работы с AutoML

h2oai / h2o-3

Public

Notifications

Fork 2k

Star 6.6k

<> Code

Issues 2.7k

Pull requests 76

Discussions

Actions

Projects

Security

Insights

master

2,557 Branches

3,833 Tags

Go to file

<> Code

maurever

GH-16063 UpliftDRF - Add error in the explain method (#16...

8d4b331 · 2 days ago

32,314 Commits

|                                 |                                                           |              |
|---------------------------------|-----------------------------------------------------------|--------------|
| .github                         | ci: Remove unwanted condition check                       | 8 months ago |
| buildSrc                        | Merge remote-tracking branch 'origin/rel-zizler'          | 3 years ago  |
| docker                          | [GH-15886] Add snyk container scan to prisma scan pipe... | 3 months ago |
| ec2                             | [PUBDEV-9081] Remove Support for Python 2.7 and 3.5 (...) | 9 months ago |
| examples/deeplearning/notebooks | PUBDEV-7232 - Remove DeepWater Estimator Function (...)   | 4 years ago  |
| gradle                          | [GH-15705] Allow GH (number) in PR description (#15707)   | 4 months ago |

About

H2O is an Open Source, Distributed, Fast & Scalable Machine Learning Platform: Deep Learning, Gradient Boosting (GBM) & XGBoost, Random Forest, Generalized Linear Modeling (GLM with Elastic Net), K-Means, PCA, Generalized Additive Models (GAM), RuleFit, Support Vector Machine (SVM), Stacked Ensembles, Automatic Machine Learning (AutoML), etc.

[h2o.ai](#)

mljar-supervised

Public

Notifications

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Actions

Projects 1

Wiki

Security

Insights

master

5 Branches

56 Tags

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<> Code

pplonski

Merge pull request #699 from DanielAvdar/master

ad55e0f · 2 weeks ago

1,150 Commits

|                   |                                                           |              |
|-------------------|-----------------------------------------------------------|--------------|
| .github/workflows | added os matrix test strategy. added trigger for PR.      | 2 weeks ago  |
| examples          | fix warning for calibration plot (#655)                   | 5 months ago |
| supervised        | optimized supervised imports statements, unused import... | 2 weeks ago  |
| tests             | optimized tests imports statements, unused imports rem... | 2 weeks ago  |
| nitignore         | Merge branch 'master' into master                         | 4 years ago  |

About

Python package for AutoML on Tabular Data with Feature Engineering, Hyper-Parameters Tuning, Explanations and Automatic Documentation

[mljar.com](#)

data-science

machine-learning

neural-network

random-forest

scikit-learn

xgboost

hyperparameter-optimization

lightgbm

Библиотеки для работы с AutoML

sb-ai-lab / LightAutoML

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Star 685

<> Code

Issues 25

Pull requests 6

Actions

Projects 1

Security

Insights

master

42 Branches

4 Tags

Go to file

Code

alexmyzhkov

Merge pull request #149 from sb-ai-lab/sswarm

7c7e3ce · 3 weeks ago

202 Commits

|             |                                              |              |
|-------------|----------------------------------------------|--------------|
| .github     | bugfix/patch-tox (#109)                      | 7 months ago |
| .gitlab     | Experiment/test exp                          | 2 years ago  |
| .jupyter    | Feature/nootebook run (#10)                  | 2 years ago  |
| docs        | Add neural networks and bug fixes (#79)      | last year    |
| examples    | sswarm + tabular utilized report in tutorial | 3 weeks ago  |
| experiments | Add test for presets (#70)                   | 7 months ago |

About

Fast and customizable framework for automatic ML model creation (AutoML)

developers.sber.ru/portal/products/lig...

python

nlp

data-science

machine-learning

regression

kaggle

binary-classification

automl

multiclass-classification

automated-machine-learning

lama

automatic-machine-learning

automl-algorithms

Скриншот: LightAutoML / [github.com](https://github.com)

# Плюсы инструментов AutoML

## Плюсы:

1. Сокращение времени на разработку
2. Сокращение времени на эксперименты
3. Переиспользование опыта лучших умов DS
4. Быстрое построение бейзлайн моделей



# Плюсы и минусы инструментов AutoML

## Плюсы:

1. Сокращение времени на разработку
2. Сокращение времени на эксперименты
3. Переиспользование опыта лучших умов DS
4. Быстрое построение бейзлайн моделей

## Минусы:

1. Недостаточно экспертизы в узких доменных областях знаний
2. Большинство моделей не интерпретируемы
3. Долгое время обучения
4. Проблемы с кастомизацией полученных моделей

# Итоги видео

- ✓ Познакомились с разнообразными библиотеками AutoML для подбора алгоритмов моделирования
- ✓ Разобрались и поняли, по каким критериям выбирать AutoML-инструменты
- ✓ Узнали, какие ограничения у использования AutoML-инструментов